Unit 9, Lesson 3: Writing a Quadratic Function from a Table

Benchmarks

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

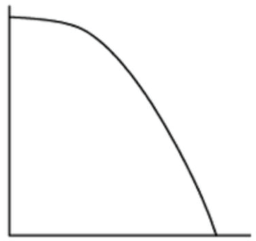
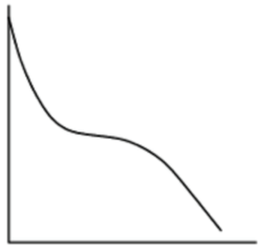
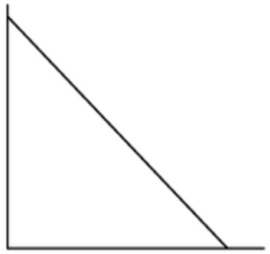
MA.912.AR.3.4 Write a quadratic function to represent the relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.

Learning Targets

- ☐ I can write a quadratic function to represent the relationship between two quantities from a table.
- ☐ I can rewrite a quadratic equation from vertex form to standard form.
- ☐ I can solve for a variable of interest.

9.3.1 Warm-Up: Words, Tables, and Graphs - Part 1

1. Match each of the following three graphs with the appropriate table of values.

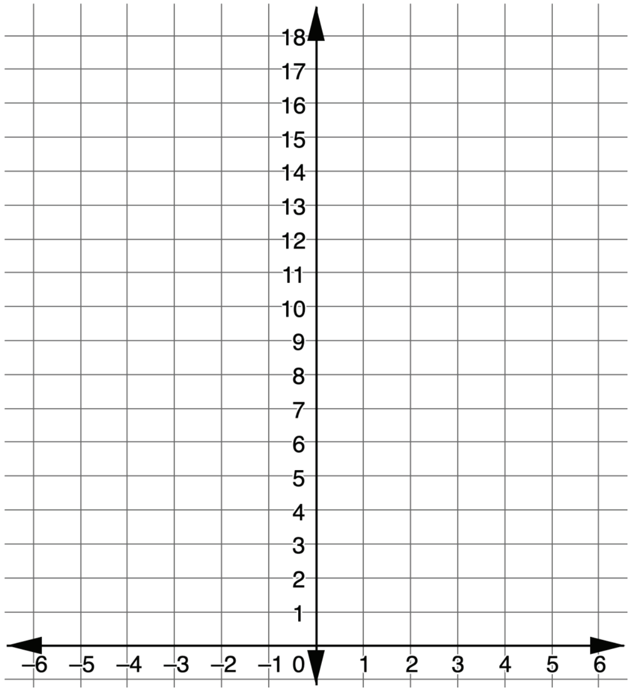
<table>
<tr><th>Graphs</th><th>Table of Values</th></tr>
<tr><td>I.
</td><td>A.
<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>400</td><td>320</td><td>240</td><td>160</td><td>80</td><td>0</td></tr></table></td></tr>
<tr><td>II.
</td><td>B.
<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>400</td><td>384</td><td>336</td><td>256</td><td>144</td><td>0</td></tr></table></td></tr>
<tr><td>III.
</td><td>C.
<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>400</td><td>253</td><td>218</td><td>216</td><td>181</td><td>34</td></tr></table></td></tr>
</table>

2. How did you use each table to determine the graph it matched to?

9.3.2 Exploration: Identifying the Vertex from a Table

1. Plot the points from the table on the graph.

x	-2	-1	0	1	2	3	4
y	17	7	1	-1	1	7	17



2. Label the vertex.

3. Discuss with your partner what change happens at the vertex in the graph. Summarize your discussion.

4. Thinking of the change that you observed in the graph, review the table and look for a similar change.

5. Discuss with your partner how the vertex can be identified from the table without plotting the points. Summarize your discussion using the key feature of increasing/decreasing.

6. On the graph, label two points that are equidistant from the vertex.

7. In the table, draw a square around the two points labeled on the graph.

8. Complete the statement.

   Two points that are equidistant from the vertex can be identified by matching points that have the same ...

9. In the table, circle two additional points that are equidistant from the vertex.

10. Discuss with your partner if there is enough information from the table to write the equation of the quadratic function in vertex form. Justify your choice by explaining your reasoning.

9.3.3 Guided Practice: Writing a Quadratic Function Using a Vertex and Point from a Table

1. The table of values for a quadratic function is provided.

x	-7	-5	-3	-1	1	3
y	-7	5	9	5	-7	-27

   a. Determine the vertex of the function.

   b. Using the vertex,, and any point, , in the table, fill in the steps for writing the equation for the quadratic function.

Step	Equation
Substitute values for the point .	
Substitute values for the vertex .	
Note: The order of the first two steps can be reversed.	
Solve for the variable .	
Write the equation by substituting in the vertex and value.	

c. Find a classmate who used a different point for . Compare your equations. Explain why they are the same or why they are different.

2. A table of values for a quadratic function is shown.

a. Identify the vertex of the quadratic function using the table.

Vertex (______ , ______)

x	y
-1	16
$\frac{1}{3}$	-8
$\frac{2}{3}$	-9
1	-8
$\frac{7}{3}$	16

b. Check your answer with your partner and discuss what strategy each of you used to identify the vertex. Correct your vertex if needed.

c. Choose a point from the table, other than the vertex, to create the equation of the quadratic in the function.

(______ , ______)

d. Determine the value of the quadratic using form.

e. Using the vertex and value, write the equation of the function in vertex form.

f. Rewrite the equation in standard form.

3. A table of values for the function is shown.

x	-11	-6	-1	4	9
$g(x)$	30	0	-10	0	30

a. Determine the equation of the function. Show your work.

b. Give the equation in both vertex and standard form.

Vertex Form	
Standard Form	

9.3.4 Your Turn!

1. Write the quadratic equation in vertex form.

x	-10	-8	-2	4	6
$f(x)$	-29	-15	3	-15	-29

Vertex	Point
(_____ , _____)	(_____ , _____)
Value of	
Vertex Form	

2. Write the quadratic equation in vertex and standard form for the quadratic function using the table.

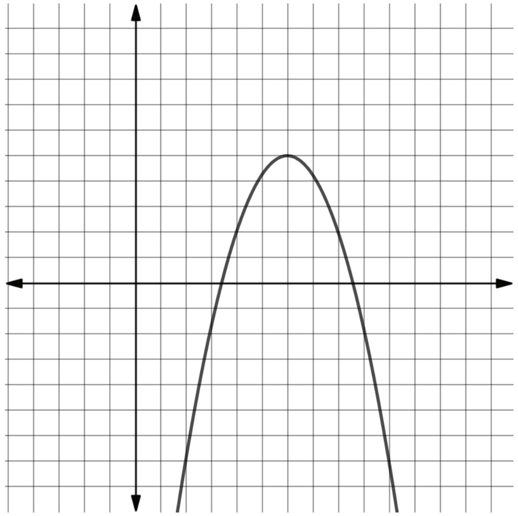
x	-5	-3	-1	1	3
$f(x)$	55	7	-9	7	55

Vertex Form	
Standard Form	

9.3.5 Wrap-Up: Ticket Out the Door

1. Write the equation for the quadratic function in vertex and standard form.

x	$f(x)$
0	-22
2	-7
4	2
6	5
8	2



Vertex Form	
Standard Form	